

Exploration Report

1. Problem Statement and Limitations of Existing hERG Modelling

hERG/KCNH2 potassium channel is one of the most important safety liabilities considered during drug development because unintended hERG blockade can interfere with cardiac repolarization and contribute to potentially serious arrhythmogenic effects. Consequently, computational prediction of hERG blockade has become an important component of early-stage drug discovery, where machine-learning models can be used to identify potentially hazardous compounds before extensive experimental testing. However, most conventional hERG machine-learning approaches primarily formulate the problem as a compound-centric prediction task. This approach can be useful for prioritizing compounds, but it implicitly treats the hERG target as a fixed entity. In other words, the model generally learns which chemical characteristics are associated with blockade of WT (Wild-type) hERG, without explicitly considering how changes in the hERG protein itself may influence the interaction. Biological complexities suggest that hERG channel is not invariant, sequence variation and mutations can alter channel properties and modify drug-channel interactions. Consequently, the same compound may not necessarily exhibit identical binding or blocking behavior against WT hERG and a mutant form of the channel. So, the real and more appropriate question that is needed to be solved is: **“How does the predicted effect of a compound change when the hERG channel itself is altered?”**. Currently, no such model exists that provides a mutant specific prediction and one of the major reasons is lack of mutant specific bioactivity datasets. Therefore, our goal not only emphasizes on building a general classification model for WT hERG but also tends to further extend and add up mutant effects utilizing the compounds of WT-hERG.

2. Work done so far

To establish a reliable foundation before addressing this problem, a WT-hERG baseline model was first developed using molecular fingerprints and physicochemical descriptors. Logistic Regression was initially evaluated as a simple baseline, followed by a Random Forest model capable of capturing nonlinear relationships between chemical features and hERG activity. The Random Forest provided the stronger baseline performance on test samples (**n=3293**), achieving **77.65% accuracy, 80.92% F1-score, 0.8652 ROC-AUC, and 0.8967 PR-AUC**. However, predictive performance alone does not indicate whether the model can be trusted for every chemical structure. Therefore, an **Applicability Domain (AD)** analysis was subsequently introduced to determine whether individual compounds occupy a chemical space sufficiently similar to that represented during model development. Using a **Tanimoto similarity threshold of >0.4**, the independent test compounds were separated into in-domain and out-of-domain groups. The model demonstrated **79.03% accuracy and 0.8759 ROC-AUC for 2,943 in-domain compounds**, compared with **66.00% accuracy and 0.7361 ROC-**

AUC for 350 out-of-domain compounds. The 13.03 percentage-point reduction in accuracy provided evidence that prediction reliability is dependent on chemical similarity to the learned chemical space. Together, these analyses establish two important foundations: the WT-hERG model provides the chemical prediction, while the applicability domain provides a measure of where that prediction is more reliable.

3. A Potential Approach to Explore and preliminary data

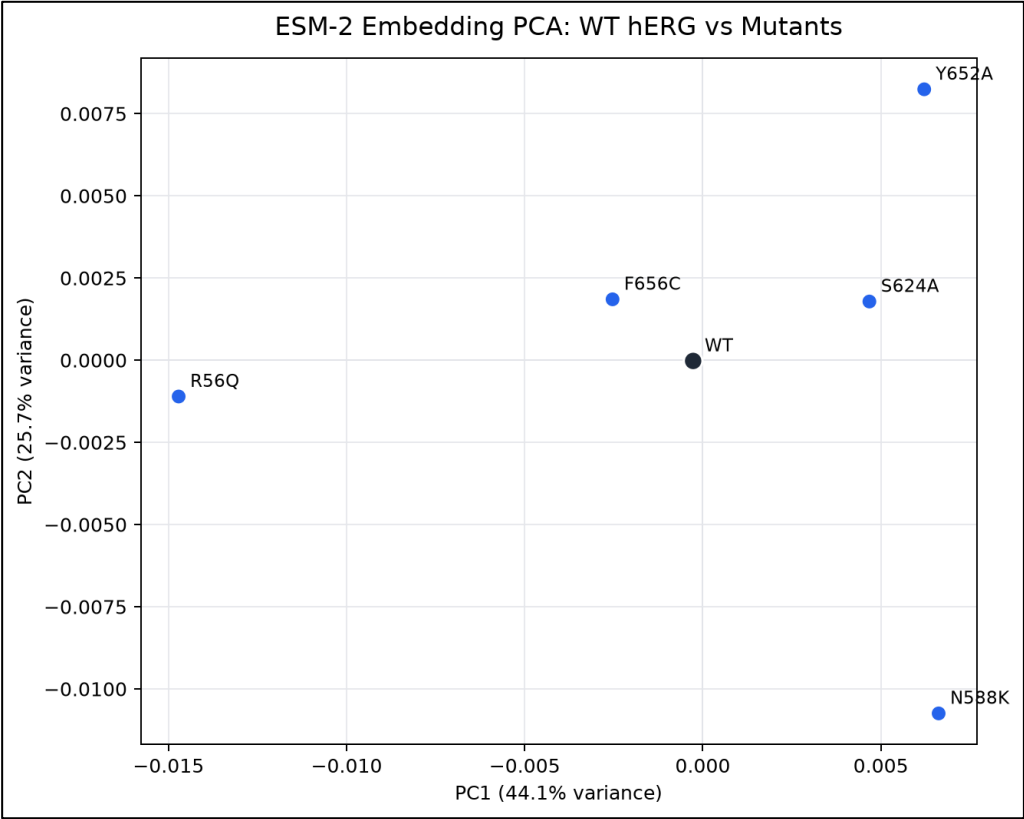
We further tend to, test a challenge of a very important missing biological dimension of mutant variations. Our hypothesis particularly asks, **“Can information about the hERG protein sequence and its mutations be incorporated into this established chemical framework to generate mutation-aware predictions, even when comprehensive mutant-specific compound activity data are unavailable?”**.

To begin testing it, we choose to first understand the variations of mutants compared to its WT using a powerful transformer-based encoding (ESM-2) from protein sequences. We generated the embeddings for 5 major variants of hERG namely, Y652A, F656C, S624A, N588K, and R56Q. Further we compared the mutant embeddings with WT to quantify how much each mutation changed the learned protein representation. We calculated measures such as Euclidean distance/cosine similarity between WT and mutant embeddings. Performed PCA visualization on the embeddings to see whether WT and the different mutants occupy distinguishable regions of the protein-embedding space. Based on the WT-distance analysis, we identified the variants showing greater or lesser representation-level changes. For example, among the values we obtained, **R56Q showed the largest WT distance (0.0168), followed by N588K (0.0149), Y652A (0.0145), S624A (0.0107), and F656C (0.0101).** Because variant-specific compound activity data were unavailable, we did not train a conventional mutant-specific classifier. Instead, we used the WT-trained chemical model as the baseline and treated the ESM-2-derived mutant information as a mutation-adjustment/proxy signal. The same compounds used for the WT model's evaluation were then passed through the framework, combining the WT predicted blocker probability with information representing the WT→mutant sequence change. This allowed us to generate hypothesis-level mutant-adjusted predictions rather than claiming experimentally validated mutant-specific activity. A significant result to showcase is **CMPD_00004 showed a marked mutation-associated reduction in predicted hERG blockade for Y652A, with blocker probability decreasing from 0.77 (WT) to 0.33, suggesting a potential reduction in compound–channel interaction following mutation.** This hypothesis demonstrated the feasibility of integrating ESM-2-derived hERG sequence information with the established WT chemical prediction model to explore potential mutation-dependent changes in hERG blockade. However, because experimentally matched compound–mutant activity data are currently limited, these predictions should be considered hypothesis-generating rather than experimentally validated mutant-specific predictions. Therefore, the next critical step is to expand the dataset with experimentally characterized compound–hERG mutant activity measurements across multiple variants and chemical scaffolds. Incorporating such data will allow the proxy

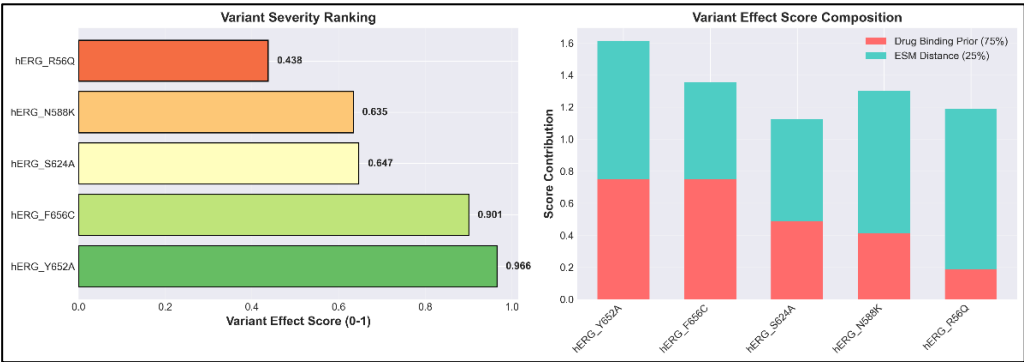
framework to evolve into a properly trained and validated mutation-specific model, improve its predictive reliability and applicability domain, and ultimately provide a stronger basis for predicting mutation-dependent hERG liability in drug discovery. The hypothesis was confirmed through experimental patch-clamp studies which have shown that hERG mutations can markedly reduce drug blockade; for example, Y652A and F656A increased the IC_{50} of verapamil by ~16- and ~20-fold, respectively, compared with WT hERG, demonstrating mutation-dependent changes in blocker sensitivity (Duan et al., 2007).

Visualizations for the tested hypothesis:

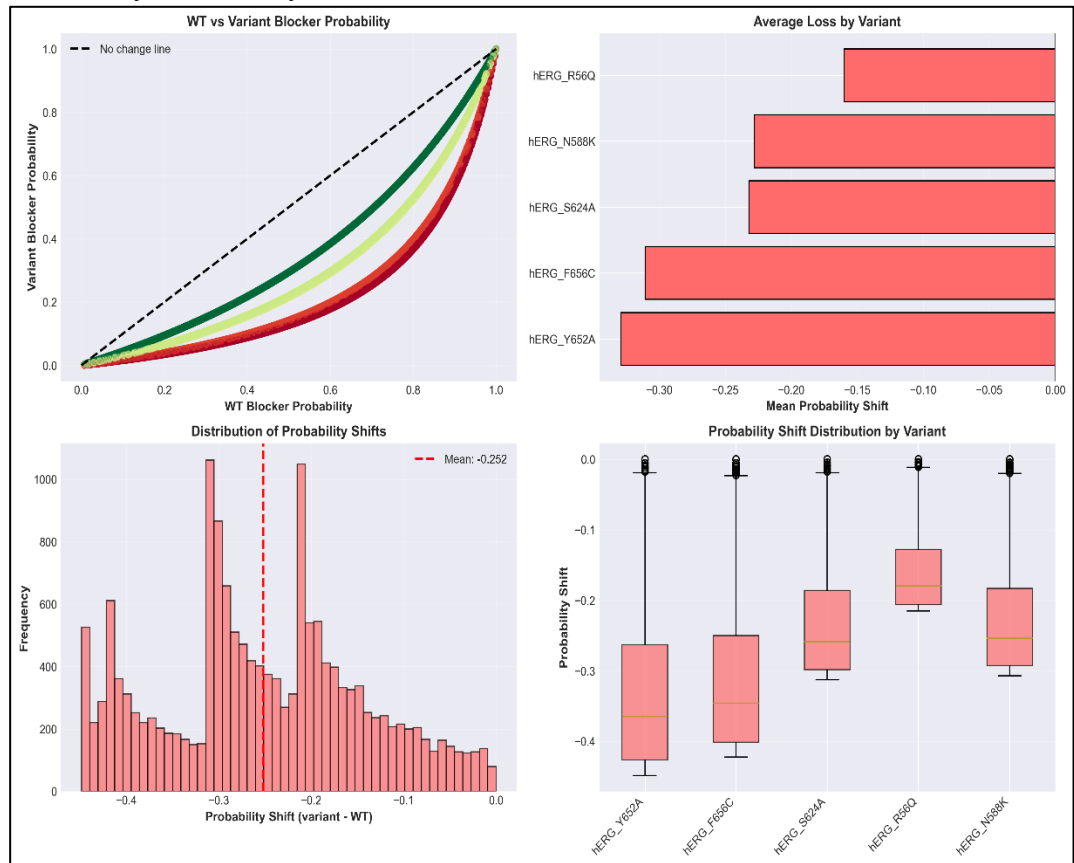
a. Embeddings visualization (PCA Plot):



b. Variant Effect Scores:



c. Probability Shift Analysis:



d. Pic50 Shift Analysis:

